MILITARY SPECIFICATION

# CHARGER, BATTERY, PP-1659( )/G

This specification is approved for use by all departments and agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers a general purpose, variable output, transformer-rectifier type battery charger designated as Charger, Battery, PP-1659( )/G. (See 6.3)

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| NN-P-71 | Pallet, Materials-handling, Wood (General Construction Requirements) |
| PPP-B-585 | Box, Wood, Wirebound |
| PPP-B-601 | Box, Wood, Cleated Plywood |
| PPP-B-621 | Box, Wood, Nailed and Lock Corner |
| PPP-B-636 | Box, Fiberboard |
| PPP-B-640 | Box, Fiberboard, Corrugated, Triple-Wall |
| PPP-F-320 | Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes |
| PPP-S-760 | Strapping, Nonmetallic (and Connectors) |
| PPP-T-76 | Tape, Pressure-sensitive Adhesive Paper, Water Resistant, (For Carton Sealing) |
| PPP-T-97 | Tape, Pressure-sensitive Adhesive, Filament Reinforced |
| QQ-S-781 | Strapping, Steel, Flat and Seals |



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation-Packaging, Methods of |
| MIL-T-152 | Treatment, Moisture And Fungus-resistant Of Communications, Electronic And Associated Electrical Equipment |
| MIL-V-173 | Varnish, Moisture And Fungus Resistant For Treatment of Communications, Electronic, And Associated Equipment |
| MIL-P-11268 | Part, Materials, and Processes Used in Electronic Communication Equipment |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finish for Ground Signal Equipment |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables for Inspection By Attributes |
| MIL-STD-108 | Definition of and Basic Requirement for Inclosure for Electric and Electronic Equipment |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-147 | Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, Runners Or Pallet Type Base |
| MIL-STD-252 | Wired Equipment, Classification Of Visual And Mechanical Defects |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics, Requirements for Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurements Of |
| MIL-STD-810 | Environmental Test Methods |

DRAWINGS

ELECTRONICS COMMAND

| | |
|---|---|
| SC-DL-359211 | Charger, Battery PP-1659( )/G |
| SC-A-678015 | Burn-in/Debugging Procedure for Charger, Battery PP-1659( )/G |

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)



Source: https://assist.dla.mil -- Downloaded: 202640-22T22:31Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First Article. - When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. - The equipment shall be constructed in accordance with Drawing SC-DL-359211.

3.2.1 Weight. - The weight of Charger, Battery, PP-1659( )/G shall not exceed 60 pounds. (See 4.14)

3.3 Parts, materials and processes; general. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Finish, protective. - Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab) semigloss enamel matching a color chip provided by the procuring agency. (See 4.4 and 6.6)

3.5 Marking. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be group 1 as described in that specification.

3.5.1 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2 Serial Number. - Each Charger, Battery, PP-1659( )/G shall be serial numbered.

3.6 Electrical Requirements

3.6.1 Input voltage. - The charger shall deliver the output voltages and current specified in 3.6.6 from the following single phase power sources:

| Volts AC (± 10%) | HERTZ (± 5%) |
|---|---|
| 115 | 50, 60 |
| 230 | 50, 60 |



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

3.6.2 Coarse Control. - The minimum voltage shall be at position 1 and maximum voltage at position 12. When tested in accordance with 4.7.1 the difference in output voltage between each coarse control tap position shall be 11.5 (± 1.0) volts D.C.

3.6.3 Fine Control. - The minimum voltage shall be at position 1 and maximum voltage at position 12. When tested in accordance with 4.7.2 the difference in output voltage between each fine control tap position shall be 1.0 (± 0.2) volts D.C.

3.6.4 Efficiency. - When tested in accordance with 4.7.3, the efficiency shall be a minimum of 70%.

3.6.5 Regulation. - When tested in accordance with 4.7.4 the maximum regulation percentage shall be 10%.

3.6.6 Output Voltages and Currents. - When tested in accordance with 4.7.5, the equipment shall be capable of providing an output current from 0 to 15 amperes DC at a voltage of 105 (± 1.0) volts D.C.

3.6.7 Operational. - The equipment shall be capable of providing an output current of fifteen amperes (± 0.5) DC and an output voltage of 105 (± 1.0) volts D.C. when tested in accordance with 4.8.1.

3.7 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.10)

3.8 Service Conditions. - The equipment shall meet the following service conditions:

3.8.1 Operation. - Continuous use for a period of seven years, under the environmental requirements specified in 3.8.2 thru 3.9, with a duty cycle of twelve hours on and twelve hours off with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.8. Approval of the preproduction sample (see 3.1) shall be considered as compliance with this requirement.

3.8.2 Temperature.

3.8.2.1 Low Temperature. - During testing as specified in 4.8.3.1, the equipment shall meet the requirements of 3.6.7.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:31Z
Check the source to verify that this is the current version before use.

3.8.2.2 High Temperature. - During testing as specified in 4.8.3.2, the equipment shall meet the requirements of 3.6.7.

3.8.3 Humidity (Moisture-Resistance). - During testing as specified in 4.8.4, the equipment shall meet the requirements of 3.6.7.

3.8.4 Altitude (Elevation.)- The equipment shall meet the requirements of 3.6.7 both during and after testing as specified in 4.8.5.

3.8.5 Vibration. - When tested in accordance with 4.8.6, the amplitude of vibration of any part, subassembly or structure of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz. Following the Vibration test the equipment shall meet the requirements of 3.6.7.

3.8.6 Electromagnetic Compatibility. - The equipment shall comply with the following emission and susceptibility limits of Table A-I-MIL-E-55301 of MIL-STD-461A, Notice 4 when tested in accordance with 4.8.7.

(a) CS02 RE 02

(b) RE 02.1

(a) The frequency range shall be 150 HKz to 65 MHz

(b) The upper frequency limit shall be 1GHz.

3.8.7 Fungus. - The equipment shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.8.8. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface. When overall tropicalization treatment of equipment is a contractual requirement, a moisture and fungus resistant coating of varnish conforming to MIL-V-173, shall be applied in accordance with MIL-T-152. The moisture-fungus resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, and solder fluxes. When tape is used on parts of equipment that are masked against treatments, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

3.9 Enclosure. - The equipment enclosure shall be of dripproof construction (15 degrees) in accordance with MIL-STD-108. After completion of the test of 4.9, there shall be no water in any pocket or bottom of the equipment. Prior to and after the inclosure tests the equipment shall meet the requirements of 3.6.7.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:31Z
Check the source to verify that this is the current version before use.

3.10 Technical literature, tools, and running spare parts. - Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.11 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268

(1) General requirements for plastic material and plastic parts.

(2) Wiring and Cabling, including:

Slack.

Protection.

Clearance.

Splicing.

Connections, general.

Grounding, general.

Shielding on wire and cable.

Soldering and brazing.

Cleaning of units.

General requirements for securing parts.

Process for welding.

Cleaning after welding.

Controls.

Self-tapping screws.

Cleaning prior to welding.



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

(b) In this specification:

Marking.

Finish.

3.12 Preconditioning.

3.12.1 Bounce. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.12.2 Burn-in. - Each equipment produced shall be subjected to the burn-in of 4.6.1 prior to Group A inspection and Preproduction inspection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

(a) First Article inspection. (See 4.3) Does not include preparation for delivery.

(b) Inspections covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12)



Source: https://assist.dla.mil -- Downloaded: 202640-22T22:31Z
Check the source to verify that this is the current version before use.

4.3 First Article. - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units. - The contractor shall furnish three (3) first article units of the complete PP-1659( )/G BATTERY CHARGER.

4.3.2 First Article Inspection. - The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified therein.

TABLE I FIRST ARTICLE INSPECTION

| INSPECTIONS | REQT | TEST PARA. | UNIT 1 | UNIT 2 | UNIT 3 |
|---|---|---|---|---|---|
| 1. Inspections covered by subsidiary documents | | 4.4 | Inspections to be performed on all units | | |
| 2. Bounce Preconditioning | 3.12.1 | 4.6 | Inspections to be performed on all units. | | |
| 3. BURN-IN | 3.12.2 | 4.6.1 | Inspections to be performed on all units. | | |
| 4. Group A Inspection | (See Table II) | | Inspections to be performed on all units. | | |
| 5. Group B Inspection | (See Table III) | | Inspections to be performed on all units. | | |
| 6. EMI | 3.8.6 | 4.8.7 | | 1 | |
| 7. Environmental | | | | | |
| Low Temperature | 3.8.2.1 | 4.8.3.1 | 1 | | |
| High Temperature | 3.8.2.2 | 4.8.3.2 | 2 | | |
| Humidity | 3.8.3 | 4.8.4 | 3 | | |
| Altitude | 3.8.4 | 4.8.5 | | 2 | |
| Vibration | 3.8.5 | 4.8.6 | | | 1 |
| Fungus | 3.8.7 | 4.8.8 | | | 2 |
| Enclosure | 3.9 | 4.9 | 4 | | |
| Weight | 3.2.1 | 4.14 | 5 | | |

SEE NOTES FOR TABLE I ON NEXT PAGE.



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

TABLE I FIRST ARTICLE INSPECTION (CONT'D)

NOTES:

(a) The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

(b) The implementation of inspections covered by 6 & 7 shall be expedited to satisfy contract delivery requirements.

(c) The numbers in the unit columns in the table covered by 6 & 7 specify the order of inspections for the indicated unit.

(d) The equipment shall be thoroughly washed, cleaned, dried and refurbished after the fungus test before proceeding with subsequent inspection.

4.4 Inspections covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

| ITEM | WHERE REQUIRED |
|---|---|
| Parts, materials, and processes-general | 3.3 |
| Finish, protective | 3.4 |
| Marking | 3.5 |

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.6.1.

4.5.1 Group A Inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspection specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of Inspection within Group A. - Group A Inspection shall be performed in an order satisfactory to the Government.



Source: https://assist.dla.mil -- Downloaded: 202640-22T22:31Z
Check the source to verify that this is the currentversion before use.

TABLE II GROUP A INSPECTION

| INSPECTION | REQT. PARA. | TEST PARA. | AQL MAJOR | AQL MINOR |
|---|---|---|---|---|
| Visual and Mechanical | 3.11 | 4.11 | 1.5 DPHU * | 6.5 DPHU |
| Electrical | | | | |
| Coarse Control | 3.6.2 | 4.7.1 | 1.5 DPHU | ** |
| Fine Control | 3.6.3 | 4.7.2 | FOR | ** |
| Efficiency | 3.6.4 | 4.7.3 | THE | ** |
| Regulation | 3.6.5 | 4.7.4 | GROUP | ** |
| Voltage Output | 3.6.6 | 4.7.5 | 1.0% | ** |

* Defects per hundred units.
** All electrical defects are major.

4.5.2 Group B inspection. - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and inspection level S-4 of Table I of MIL-STD-105.

TABLE III - GROUP B INSPECTION

| INSPECTION | REQUIREMENT PARAGRAPH | TEST PARAGRAPH | AQL |
|---|---|---|---|
| Interchangeability | 3.7 | 4.10 | 6.5% |

4.5.3 Group C inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. - One sample shall be selected at random for each group of inspections in Table IV from every 200 or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:31Z
Check the source to verify that this is the current version before use.

4.5.3.2 Order of inspection within Group C - Group C inspection shall be performed in the order shown in each group in Table IV.

4.5.3.3 Group C Failures. - Actions required relative to Group C failures shall be as specified in the contract or purchase order.

TABLE IV GROUP C INSPECTION

| INSPECTION | REQ. PARA. | INSP. PARA. |
|---|---|---|
| GROUP I | | |
| Low Temperature | 3.8.2.1 | 4.8.3.1 |
| High Temperature | 3.8.2.2 | 4.8.3.2 |
| Altitude | 3.8.4 | 4.8.5 |
| Humidity | 3.8.3 | 4.8.4 |
| GROUP II | | |
| Vibration | 3.8.5 | 4.8.6 |
| Fungus | 3.8.7 | 4.8.8 |
| GROUP III | | |
| Enclosure | 3.9 | 4.9 |
| EMI | 3.8.6 | 4.8.7 |
| Weight | 3.2.1 | 4.14 |

4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and pass both Group B and Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are subject to and pass Group A inspection.

4.6 Bounce Preconditioning. - Each unit of each lot shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The Package Tester, shafts in place, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection. (See 3.12.1)



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the currentversion before use.

4.6.1 Burn-in. - Each unit of each lot shall be subjected to a burn-in period, after the Bounce Preconditioning, in accordance with SC-A-678015. (See 3.12.2)

4.7 Electrical Test.

4.7.1 Coarse Control. - With an input of 115 VAC, 60 Hertz, no load, the fine control shall be set at position 1. When the "coarse" control is rotated from position 1 through 12, the DC voltage difference between each position shall meet the requirements of 3.6.2.

4.7.2 Fine Control. - With an input of 115 VAC, 60 Hertz, no load, the coarse control shall be set at position 1. When the "fine" control is rotated from position 1 through 12, the DC voltage difference between each position shall meet the requirements of 3.6.3.

4.7.3 Efficiency. - The battery charger shall be operated at full load (15 amps-resistive) with an input of 115 VAC, 60 Hertz. The control tap switches shall be set to positions 12-12. The efficiency shall be calculated using the following formula and meet the requirements of 3.6.4.

$$\text{Efficiency \%} = \frac{(\text{Output volts})\ (\text{Output amps DC}) \times 10^{2}}{\text{Input Watts}}$$

4.7.4 Regulation. - The battery charger shall be operated with an input of 115 VAC, 60 Hertz, no load. The control tap switches shall be set to positions 12-12. A 15 ampere resistive load shall be applied and the input adjusted to maintain 115 VAC and 60 Hertz. The regulation shall be calculated using the following formula and meet the requirements of 3.6.5.

$$\text{Regulation \%} = \frac{(\text{Output VDC, no load} - \text{Output VDC, 15A load}) \times 10^{2}}{\text{Output VDC, no load}}$$

4.7.5 Output voltage and current. - The output voltage and current shall meet the requirements of 3.6.6 when the input voltage, and frequency is varied as specified below:

| Input Voltage (AC) | Freq. (HERTZ) |
|---|---|
| 103.5 | 47.5, 63 |
| 115.0 | 47.5, 60, 63 |
| 126.5 | 47.5, 63 |
| 207.5 | 47.5, 63 |
| 230.0 | 47.5, 60, 63 |
| 253.0 | 47.5, 63 |



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

4.8 Service Condition Tests.

4.8.1 Operational. - The battery charger shall supply an output voltage in accordance with 3.6.7 under the following conditions for each load specified below:

| Input Voltage (AC) | Frequency (hertz) | Res. Load (amps) |
|---|---|---|
| 115 | 60 | 2, 15 |
| 230 | 60 | 2, 15 |

NOTE: The "coarse" control and "fine" control, during the service condition testing, shall remain in the same positions required to meet 105 VDC out, with a 230 VAC, 60 hertz input, and a 15 amp resistive load.

4.8.2 Electrical requirements during service condition tests. - With an input of 230 VAC, 60 hertz, full load, the battery charger shall meet the performance requirements specified. (See Note 4.8.1)

4.8.3 Temperature Test. -

4.8.3.1 Low Temperature. - The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -80°F and the operating temperature shall be - 40°F. Prior to testing the equipment shall meet full specification performance for measurements specified in 3.6.7. (See 3.8.2.1)

4.8.3.2 High Temperature. - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 150°F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.6.7. (See 3.8.2.2)

4.8.4 Humidity (moisture-resistance) test. - The Charger, Battery PP-1659( )/G shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for measurements specified in 3.6.7. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements; and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. (See 3.8.3)



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

4.8.5 Altitude (elevation) test. - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.6.7. (See 3.8.4)

4.8.6 Bounce (vibration) test. - The equipment shall be tested in accordance with Method 514, Procedure XI, of MIL-STD-810, modified as follows: Ref Par. 4.11.2.4 - "Performance of test" - Delete the fifth sentence in its entirety (At the end of each 1/2 hour period, etc.) and replace with the following:

"The test item shall be positioned with the skid mounts on the tester bed, and the right side facing the backboard. At the end of the first hour, the test item shall be rotated so that the rear of the unit faces the backboard. At the end of the second hour, the test item shall be rotated so that the left side of the unit faces the backboard. The input cable shall be secured to the unit in a manner to provide minimum contact with the side-board and backboard of the tester during the test operation." (See 3.8.5)

4.8.7 Electromagnetic Compatibility. - The equipment shall be tested for compliance with the requirements of 3.8.6. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3. Test method CS02 shall be performed utilizing procedure B, LISN Method.

The battery charger shall be operated using a 115 VAC, 60 Hz input and a 105 VDC 15 ampere resistive load output.

4.8.8 Fungus test. - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required. (See 3.8.7)

4.9 Enclosure Test. - The battery charger, in its normal operating position, shall be subjected to the dripproof(15 degrees) test specified in MIL-STD-108 to determine compliance with 3.9.

4.10 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when dimensions affect mating of parts.



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.11 Visual and mechanical inspection. - The equipment shall be examined for defects listed in MIL-STD-252. (See 3.11)

4.12 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.13 Rough handling test (preparation for delivery). - When rough handling test in accordance with MIL-P-116 is required by contract (see 6.2(d)2c), the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling.

| Insp. Para. | Measurement |
|---|---|
| 4.8.1 | Output |

4.14 Weight. - The equipment shall be weighed to determine compliance with 3.2.1.

5. PREPARATION FOR DELIVERY. -

5.1 Preservation and Packaging. - Preservation and packaging shall be level A or C as specified (See 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. - Each Charger, Battery PP-1659( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each Charger, Battery PP-1659( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation Application. - None required.



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Charger, Battery PP-1659( )/G. - Each Charger, Battery PP-1659( )/G shall be packaged method III as follows: Cushion each Charger, Battery PP-1659( )/G on all surfaces with cells or pads, or both, fabricated from fiberboard conforming to PPP-F-320, CF, W5c, weather resistant, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close fitting fiberboard box conforming to PPP-B-636, W5c, weather resistant. Place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. - Charger, Battery PP-1659( )/G shall be packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the con tainer(s) or its contents. (See 6(b)).

5.2.1 Level A.

5.2.1.1 Palletized load. - A quantity of Charger, Battery PP-1659( )/G, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class-weather resistant, W5s, or V3c. The load shall be "bonded" to the pallet by strapping, conforming to QQ-S-781, type 1, finish A or shrink film.

5.2.1.2 Less than palletized load. - When quantities perdestination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

5.2.1.2 (Cont'd) or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. - A quantity of Charger, Battery PP-1659( )/G, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Charger, Battery PP-1659( )/G, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. - A quantity of Charger, Battery PP-1659( )/G packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Charger Battery PP-1659( )/G, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2, except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:31Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended Use. - Charger, Battery, PP-1659( )/G is intended to provide dc current to charge one or more secondary batteries.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendments thereto.

(b) Level of packaging and level of packing required for shipment. (See Section 5)

(c) When rough handling and functional tests are required.

(d) Preproduction inspection:

(1) Three sample units of each item cited in section 1 are generally required so that lengthy and potentially damaging environmental tests can be completed on one sample unit while complete performance measurements can be made on the remaining sample units. (See 3.1)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon. (See 4.13)

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.10)

(h) Quantity of tools and running spare parts required. (See 3.10)

(i) Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.8.7)

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design for example: PP-1659W/G. The contractor should apply for nomenclature (Type Designation) in accordance with the applicable clause in the contract.



Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

6.4 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of the corrective action. (See 4.5.3.3)

6.6 Color. - The color chip furnished by the procuring agency (3.4) will match color chip no. X-24087 and may be obtained upon request to Commanding General, Coating and Chemical Laboratory, Aberdeen Proving Ground, Maryland.

6.7 Location of operational inspection. - It is desirable that the operational inspection (4.8.1) be performed at a location that will minimize handling (which might cause damage to the shipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging. (See 4.5.1)

6.8 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, construction methods, etc.

6.8.1 Temperature. -

(a) Operating: Ambient temperature in the range of + 155°F to - 40°F. (The high temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature not to exceed 6 hours, at any one time.

(b) Nonoperating: Exposure in the range of + 160°F to - 80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 6 hours, at any one time.

6.8.2 Relative humidity. - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.



Source: https://assist.dla.mil -- Downloaded: 20260-22T22:31Z
Check the source to verify that this is the current version before use.

6.8.3 Altitude (elevation). -

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

CUSTODIAN:

ARMY-EL

PREPARING ACTIVITY

Army-EL

Project Nr. 6130-A135



☆ U. S. GOVERNMENT PRINTING OFFICE: 1973-713-159/515

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:31Z
Check the source to verify that this is the current version before use.

# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION CHARGER, BATTERY, PP-1659( )/G MIL-C-55158B(EL)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 20261O-22T22:31Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL- RD-ED
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:31Z
Check the source to verify that this is the current version before use.